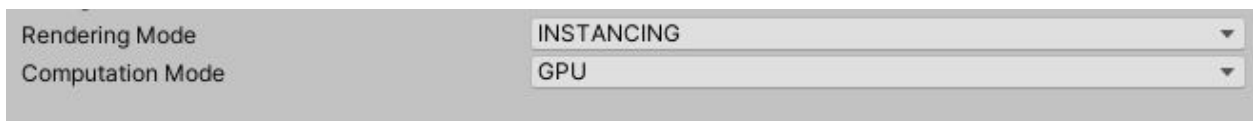




Drag and drop the FishFlockController2.cs script to any object on your scene, and configure each property through the inspector. Each property is described with a tooltip, just put the mouse cursor over the property and it will show a description of it. The tool supports unity's built-in renderer and both Universal and HD rendering pipelines.

The Script

The script contains two different modes for rendering and computing the fishes behaviours.



You have an INSTANCING and GAME OBJECT modes for rendering, the difference is that in the instancing mode it will render the fishes using the instancing feature and you will not have any fishes as game object, and in the game object one you have each fish being its own game object with a mesh renderer.

In the Computation Mode case, you have a CPU, CPU_AND_ONE_THREAD and GPU. The CPU calculates the fishes using normal code and the CPU_AND_ONE_THREAD will use a parallel thread to calculate it, but the GPU mode uses a Compute Shader to calculate the flocking behaviour.

Demos

On the demo's folder, there are demos like:

- FishFlock_Sample: This demo will show a lot of variations for each configuration you can do with the tool, don't forget to disable some of the controllers if you don't have a mid-high processor (there are multiple controllers at this scene simulating fishes).
- FishFlock_FollowRandomPoints: This demo will show you how the fishes behave when they follow a segment of generated points on the scene, inside the bounds of the controller.
- Collision Avoidance: This will show a basic example of how the fishes avoid colliders in the scene that you can drag and drop on the inspector.

Shaders

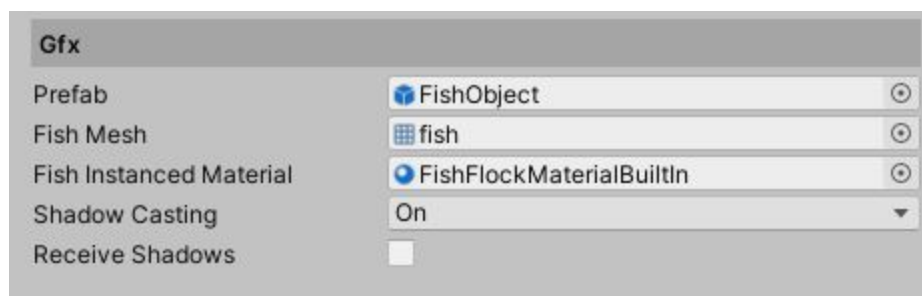
There are 3 shaders (Built-In Renderer, URP and HDRP versions) included that apply a simple effect on the fish model that simulates a fish swimming, it has these properties on its material:

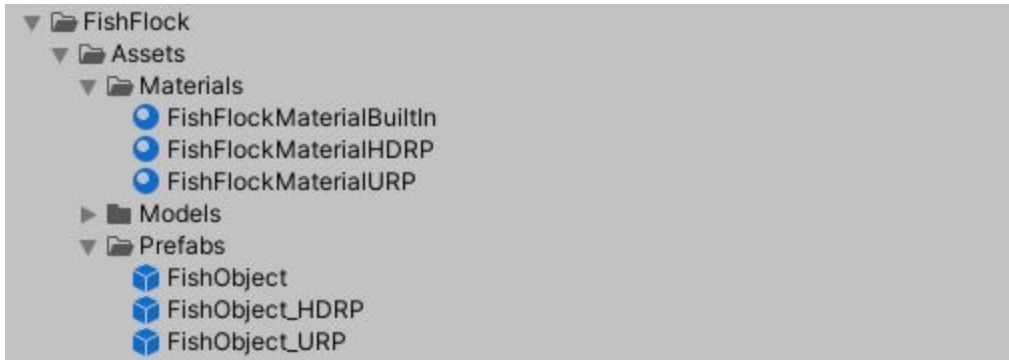
- Animation Speed: The speed that the fish will animate to simulate the swimming.
- Scale: How much the fish will animate, the lower the value is less movements it does.
- Yaw: The usual swimming effect where the fish makes a "ping-pong" effect with his head and its body follows.
- Roll: The movement scale the fish does with the lower part of its body to the left/right.

These shaders are vital for the instancing and the compute shader that the GPU uses, you can work and edit them or you can use your own custom shaders using just the CPU computation and GAMEOBJECT rendering.

Switching between render pipeline resources

In this area you can switch the resources to render with the current selected render pipeline. On the assets folder that comes with the asset it contains materials and prefabs ready to use.





Contact

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Thanks for purchasing!

Good luck!